Percentages: Change, Interest & Financial Maths

Quick reference sheet · GCSE Maths, Number · keep handy for revision

1. The multiplier method

Increase by X% → multiplier = 1 + (X ÷ 100). e.g. increase by 15% → **1.15**

Decrease by X% → multiplier = 1 – (X ÷ 100). e.g. decrease by 20% → **0.80**

Apply it by multiplying once: new amount = original × multiplier.

Change	Multiplier
Increase by 8%	1.08
Increase by 50%	1.50
Decrease by 12%	0.88
Decrease by 35%	0.65

2. Reverse percentages

Given the amount *after* a change, find the original (100%) amount by **dividing by the multiplier** — do not add or subtract the percentage from the final amount.

£69 after a 15% increase → 69 ÷ 1.15 = **£60** (original)

£120 after a 20% decrease → 120 ÷ 0.8 = **£150** (original)

3. Simple interest

I = P × R × T ÷ 100 — P = principal, R = rate (%), T = time (years). Adds the same amount each year.

£500 at 4% for 3 years → I = 500 × 4 × 3 ÷ 100 = **£60**

4. Compound interest

Amount = P × (multiplier)^{T} — interest is earned on interest, so it grows faster than simple interest.

£500 at 4% for 3 years → 500 × 1.04^{3} = **£562.43**

Year	Simple total (£500 @ 4%)	Compound total (£500 @ 4%)
1	520.00	520.00
2	540.00	540.80

3	560.00	562.43

5. Before you start: what is 100%?

For "increase/decrease by X%" questions, 100% is the **original** amount. For reverse percentage questions, the amount given is **not** 100% — it is the multiplier percentage, so divide by the multiplier to get back to the original.